pwn.college: File Struct

Reading and Writing with FILE

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FILE struct - Buffer Pointers

```
int _flags;            /* High-order word is _IO_MAGIC; rest is flags. */

char *_IO_read_ptr;  /* Current read pointer */
char *_IO_read_end;  /* End of get area. */
char *_IO_read_base; /* Start of putback+get area. */
char *_IO_write_base;    /* Start of put area. */
char *_IO_write_ptr; /* Current put pointer. */
char *_IO_write_end; /* End of put area. */
char *_IO_buf_base;  /* Start of reserve area. */
char *_IO_buf_end;   /* End of reserve area. */

int _fileno;
```

https://elixir.bootlin.com/glibc/glibc-2.31/source/libio/bits/types/struct_FILE.h#L49

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FILE struct - Flag Useful Values

```
#define _IO_MAGIC           0xFBAD0000 /* Magic number */

#define _IO_UNBUFFERED          0x0002

#define _IO_NO_READS            0x0004 /* Reading not allowed.  */

#define _IO_NO_WRITES           0x0008 /* Writing not allowed.  */

#define _IO_CURRENTLY_PUTTING 0x0800

#define _IO_IS_APPENDING        0x1000
```

https://elixir.bootlin.com/glibc/glibc-2.31/source/libio/libio.h#L62

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Gaining arbitrary read & write

Stream libraries

- Read from a file to memory
- Write memory out to a file

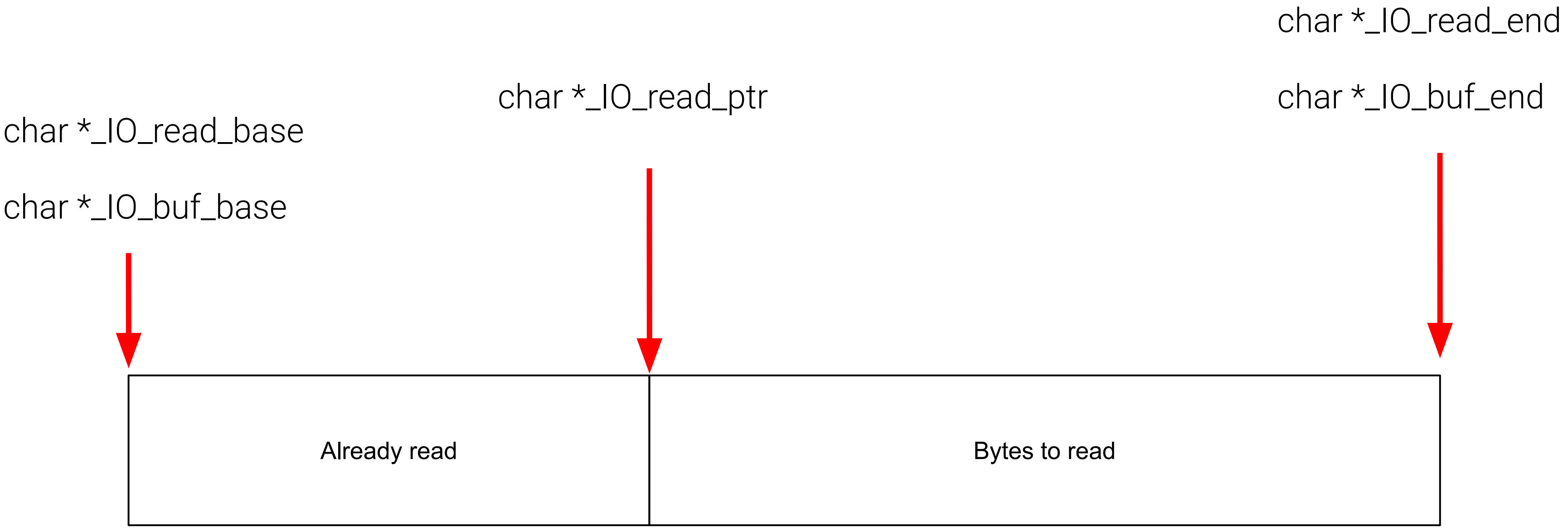
Overwriting the buffer pointers can control where the read/write occurs!

FILE struct - Read() ing to Arbitrary Memory

Requirements

- Set flag value
- Set `read_ptr` = `read_end`
- Set `buf_base` to address to write
- Set `buf_end` to address to write + length
- `buf_end - buf_base` >= number of bytes to read

FILE struct - Read() ing to Arbitrary Memory



*File must be opened for read!

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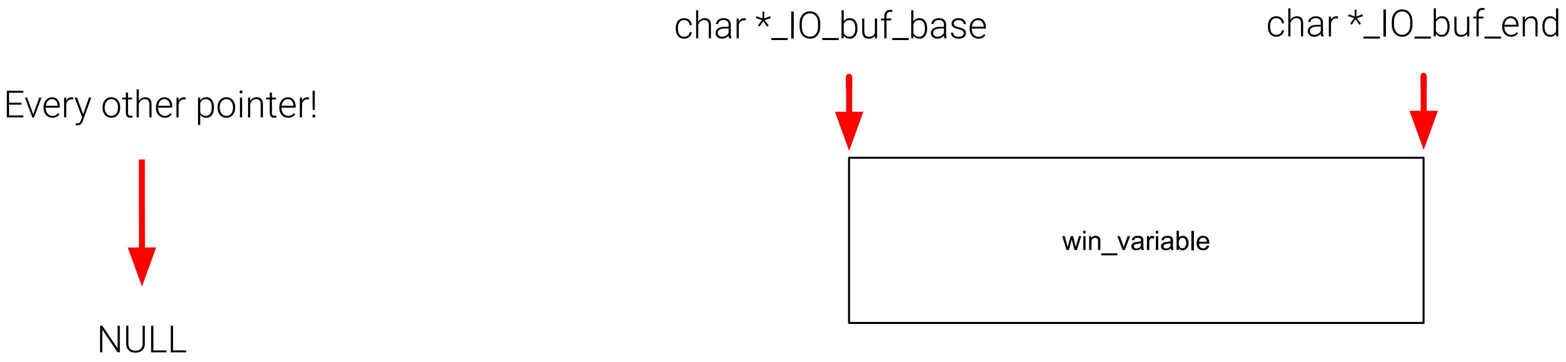

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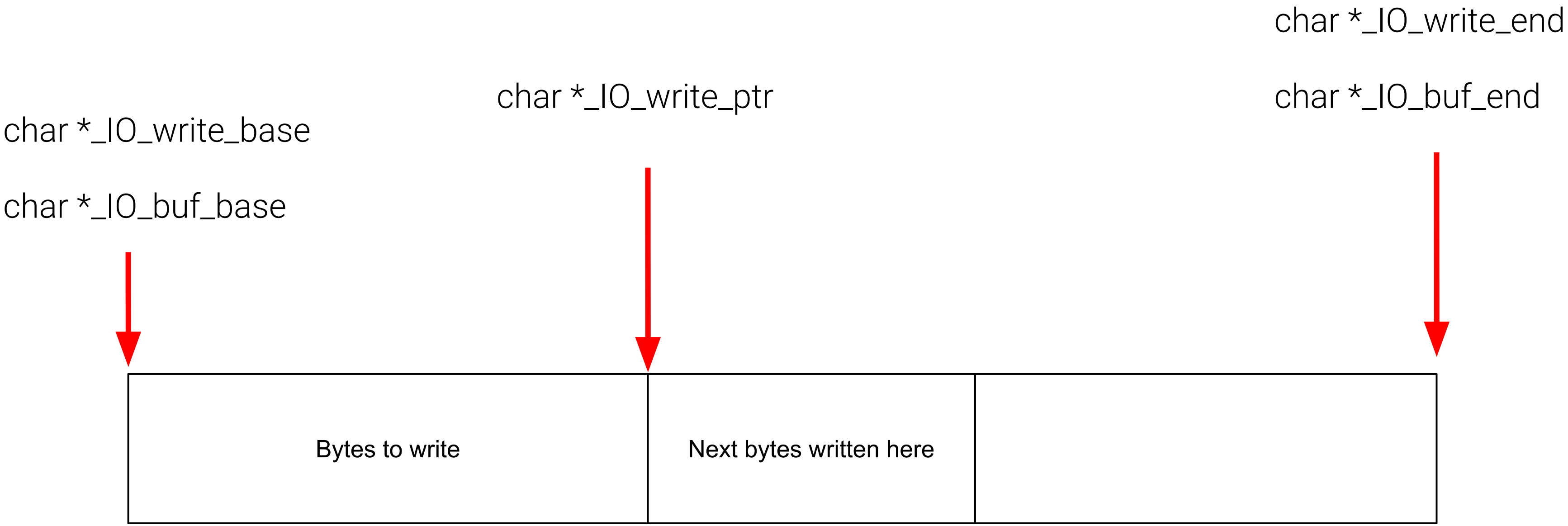
FILE struct - Write() ing from Arbitrary Memory

Requirements

- Set `flag` value
- Set `write_base` to memory to write
- Set `write_ptr` to address to write + length
- Set `read_end` = `write_base`

FILE struct - Write() ing from Arbitrary Memory



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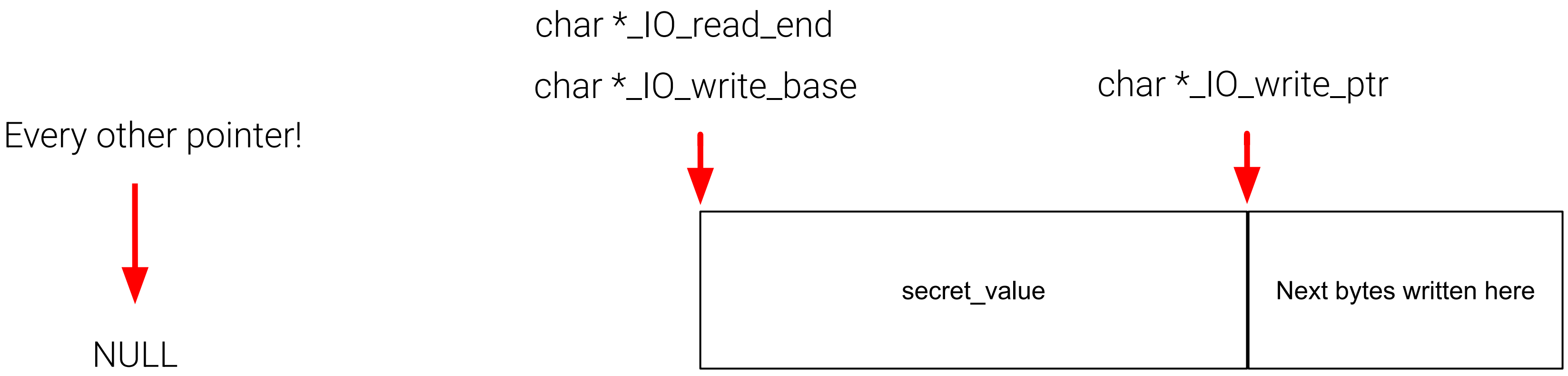

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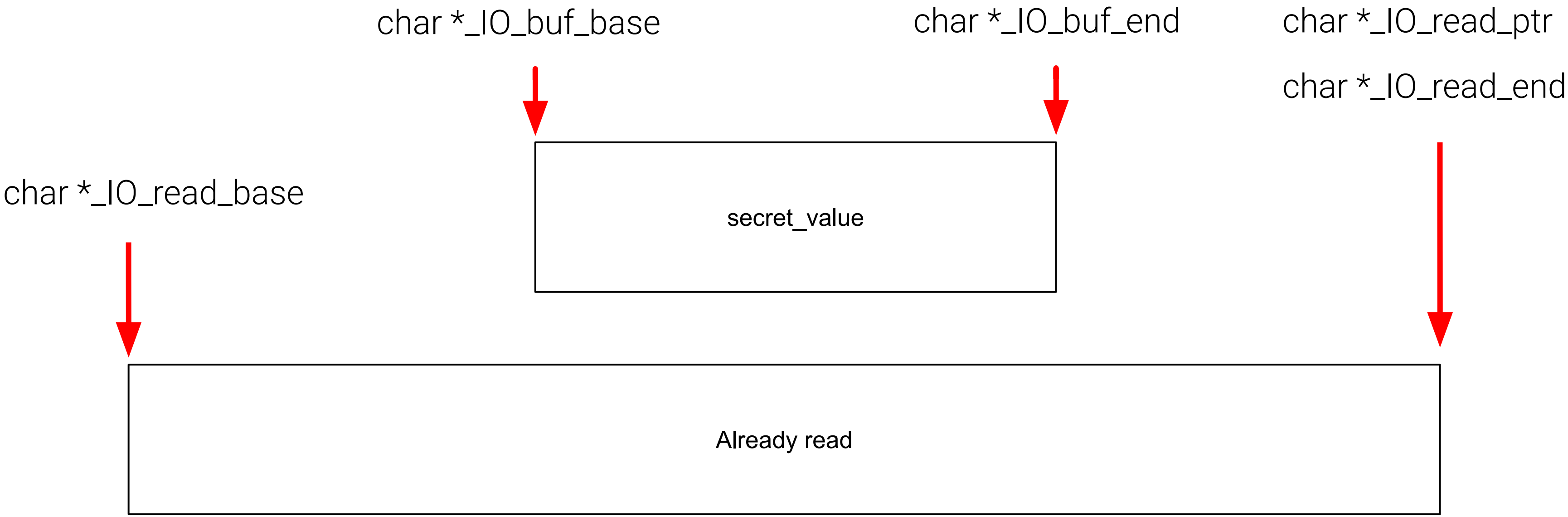
FILE struct - Read() ing to Memory



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